

Can Weight-Space Encoders Predict Generalization Gap?

A Controlled Study of Equivariant Architectures

Anonymous
Anonymous Institution
anonymous@anonymous.edu

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Abstract

Predicting how much a trained neural network has overfit — its generalization gap — is more tractable from weight tensors than predicting its test accuracy outright. Motivated by this counterintuitive asymmetry, we conduct the first controlled comparison of equivariant and non-equivariant weight-space encoders on generalization gap as the primary prediction target, using a dual-target design on the Unterthiner CIFAR-10 CNN zoo. We find that gap is learnable at Spearman $r > 0.5$ with standard encoders, and that architecture specificity matters: only Neural Functional Transformers, with their cross-layer attention, improve over the flat baseline on gap ($r = 0.575$), while within-layer and graph-structured equivariant architectures underperform it. Most strikingly, a partial Spearman analysis reveals that NFT gap predictions carry substantial information about true gap that is independent of test accuracy — confirming that generalization gap and test accuracy encode distinct signals in weight space. We report both the positive findings and a null result on the target-specificity mechanism hypothesis, with identified confounders and proposed corrective experiments, providing a transparent empirical foundation for gap-targeted weight-space learning.

1 Introduction

A model’s weight tensors predict its generalization gap — how much training accuracy exceeds test accuracy at convergence — better than they predict its test accuracy outright. In the Unterthiner CIFAR-10 CNN zoo, a simple position-indexed MLP achieves Spearman rank correlation $r = 0.56$ on generalization gap prediction but only $r = 0.28$ on test accuracy prediction, despite receiving the identical weight tensor as input. The harder prediction target, as it turns out, is the more legible one.

This asymmetry is not merely a curiosity. Generalization gap ($\text{train_acc} - \text{test_acc}$ at convergence) is the direct quantitative signature of overfitting — the quantity that determines whether a model has memorized its training data or extracted transferable patterns. For practitioners managing large model zoos, identifying low-gap models without held-out test evaluation would dramatically reduce the cost of model selection. For theorists, knowing that weight tensors encode an accessible gap signal raises a structural question: what is the geometric nature of overfitting in weight space, and which architectural inductive biases are best suited to extract it?

The gap in existing work. A rich literature has established that weight-space encoders can predict test accuracy with Spearman correlations as high as $r \approx 0.9$. Unterthiner et al. [2020] introduced the model zoo benchmark and flat MLP baseline. Navon et al. [2023] demonstrated that permutation-equivariant Deep Weight Space networks (DWS) substantially outperform flat baselines on test accuracy. Zhou et al. [2023] showed that Neural Functional Transformers (NFT) — cross-layer attention architectures respecting weight-space symmetries — achieve competitive or superior performance. Kofinas et al. [2024] extended this with graph neural network encoders over the neural network graph. Critically, **all these works benchmark exclusively on test accuracy prediction**. Generalization gap is computable from the same zoo metadata yet has never been treated as the primary prediction target in this line of work — and no study has examined whether encoder architecture advantage is target-specific (gap vs. test_acc) or uniform across targets.

Our approach. We conduct the first controlled comparison of equivariant vs. non-equivariant weight encoders on generalization gap as the primary prediction target. The central methodological innovation is a dual-target experimental design: the same encoder trained on the same zoo with the same hyperparameter budget, evaluated on both gap and test accuracy prediction. This design isolates target-specificity from encoder-specificity. We additionally introduce a partial Spearman analysis — the first in this literature — that tests whether gap predictions contain information about true gap that is statistically independent of test accuracy rank.

Key finding. NFT’s cross-layer attention captures gap-specific overfitting structure in weight tensors that is statistically independent of test accuracy: $\text{partial Spearman}(\text{NFT_gap_pred}, \text{true_gap} \mid \text{true_test_acc}) = 0.73$, $p = 1.6 \times 10^{-167}$. This result — substantially exceeding NFT’s direct gap Spearman of 0.57 — indicates that the gap-specific component of NFT’s predictions is stronger than the combined signal, and that generalization gap and test accuracy occupy distinct regions of the weight-space information landscape.

Contributions. We make four contributions:

1. **Gap learnability (existence).** We establish that generalization gap is predictable from weight tensors at Spearman $r > 0.5$ using both position-indexed (FlatMLP $r = 0.5567$) and equivariant (DWSNet $r = 0.5104$) encoders, with a confirmed A1 audit showing gap is not trivially derivable from test accuracy ($\text{Spearman}(\text{gap}, -\text{test_acc}) = -0.142$).
2. **Architecture specificity.** Among the four tested encoders, NFT achieves the highest gap Spearman ($r = 0.5752$, 95% CI: $[0.5339, 0.6158]$). FlatMLP’s 95% CI from the same run is $[0.4850, 0.5801]$; the two intervals overlap at the margin ($[0.5339, 0.5801]$), but NFT’s lower bound exceeds FlatMLP’s point estimate (0.5330), providing directional statistical evidence of advantage. DWSNet ($r = 0.4881$) and GNN ($r = 0.3747$) both underperform FlatMLP on gap — showing that equivariance alone is insufficient; cross-layer reasoning matters.
3. **Gap-specific signal independence (P3).** NFT gap predictions contain substantial information about true generalization gap that is independent of test accuracy (partial Spearman $r = 0.73$, $p \approx 0$), establishing gap as a genuinely distinct weight-space prediction target.
4. **Documented null result (Δ mechanism).** The differential advantage hypothesis — that equivariant encoders show a larger Spearman improvement on gap than on test_acc ($\Delta > 0.02$) — is not confirmed under our experimental conditions. We identify a plausible confounder (anomalously low FlatMLP test_acc in our zoo, $r = 0.28$ vs. literature ≈ 0.85) and propose corrected future experiments.

We organize the paper as follows. Section 2 surveys weight-space encoder and generalization prediction literature, positioning our dual-target formulation against prior work. Section 3 describes the experimental methodology including the dual-target design and partial Spearman analysis. Section 4 presents experimental setup. Section 5 presents results. Section 6 discusses mechanism interpretation, limitations, and broader implications. Section 7 concludes.

2 Related Work

We organize related work around two threads that our work bridges: (A) weight-space encoders for model property prediction, and (B) generalization gap characterization. Our contribution lies at the intersection: we apply learned weight-space encoders to gap prediction and introduce target-specificity analysis.

2.1 Weight-Space Encoders for Model Property Prediction

Flat baselines and the model zoo benchmark. Unterthiner et al. [2020] introduced the foundational benchmark: a zoo of $\sim 14,000$ small CNNs trained on CIFAR-10 with varying hyperparameters, with test accuracy labels. A position-indexed MLP operating on flattened, sorted weight tensors achieves Spearman $r \approx 0.85$ on test accuracy prediction, establishing the competitive bar. Eilertsen et al. [2020] complemented this with hand-crafted weight statistics (spectral norms, layer means, singular value distributions), showing

that informative properties can be extracted without learned encoders. Both works treat test accuracy as the sole prediction target — a framing that all subsequent encoder work has inherited.

Permutation-equivariant encoders. The core insight motivating equivariant architectures is that neural network weights are only defined up to permutation of hidden neurons: any permutation of neurons within a layer, with corresponding column/row permutations of adjacent weight matrices, yields a functionally equivalent network. Flat MLPs break this symmetry by sorting weights — a preprocessing heuristic that may introduce spurious features. Navon et al. [2023] proposed Deep Weight Space (DWS) networks: a family of permutation-equivariant layers that compute statistics shared across neuron permutation equivalence classes. DWS achieves Spearman $r \approx 0.9$ on Unterthiner test accuracy prediction, substantially outperforming the flat baseline. Crucially, DWS enforces within-layer equivariance — each layer’s neurons are treated as an interchangeable set — but does not explicitly model cross-layer interactions.

Cross-layer attention. Zhou et al. [2023] introduced Neural Functional Transformers (NFT): a transformer-based architecture that attends across the full sequence of weight matrices while respecting the symmetry group of the neural network. By operating on sequences of weight rows and columns across layers, NFT captures inter-layer weight co-variation — a qualitatively different inductive bias from DWS’s within-layer averaging. NFT achieves competitive performance on test accuracy prediction and enables symmetry-aware fine-tuning of neural networks. Our work is the first to apply NFT to gap prediction and to demonstrate that its cross-layer attention specifically captures gap-independent signal.

Graph neural network encoders. Kofinas et al. [2024] represent neural networks as graphs — neurons as nodes, weights as edges — and apply standard graph neural networks to produce equivariant weight embeddings. This representation is theoretically appealing but imposes a rigid graph connectivity structure that may over-constrain the representation for distributed properties like generalization gap. In our experiments, GNN achieves the lowest gap Spearman ($r = 0.3747$) among all tested encoders, below even the flat baseline — suggesting the graph connectivity structure does not align well with the distributed nature of gap signal.

Universal and cross-architecture encoders. Trabucco et al. [2024] proposed Universal Neural Functionals (UNFN) designed to process networks of varying architectures without retraining. Schürholt et al. [2022] developed hyper-representations via self-supervised pretraining on model zoos, enabling cross-architecture transfer. While both works are relevant to the long-term goal of architecture-agnostic gap prediction, our study is restricted to a single architecture family (Unterthiner CIFAR-10 small CNNs) to ensure clean controlled comparison — a prerequisite for causal attribution of encoder-specific effects.

The gap blind spot. A striking commonality across all encoder papers above is exclusive focus on test accuracy as the prediction target. Generalization gap — while computable from the same model zoo metadata — has not been systematically studied as a prediction objective, and target-specificity (whether encoder advantages transfer from test_acc to gap) has never been examined. Our work addresses this gap directly.

2.2 Generalization Gap Characterization

Complexity measures. A parallel literature characterizes generalization gap using hand-crafted complexity measures: sharpness of the loss landscape, margin-based bounds, PAC-Bayes flatness measures. Jiang et al. [2020] conducted a comprehensive empirical study comparing 40 complexity measures as predictors of generalization gap across model zoos, finding that no single measure dominates across settings. Their work establishes that generalization gap is a challenging, multi-dimensional property. Our contribution adds learned weight-space encoders to this comparison family, showing that NFT achieves Spearman $r > 0.5$ — comparable to the better complexity measures in their study, without requiring problem-specific feature engineering.

Theoretical connections. PAC-Bayes generalization bounds are phrased in terms of the KL divergence between learned and prior weight distributions — a quantity that is computed over the full weight tensor and is permutation-invariant by construction. This theoretical connection motivates the hypothesis that permutation-equivariant encoders, which compute orbit-averaged weight statistics, may have an inductive bias aligned with gap signal. Our experimental findings partially support this (NFT captures genuine gap-specific signal) while also revealing that the connection is not simply “equivariance $\Rightarrow$ gap advantage” — architecture specifics (cross-layer vs. within-layer) matter.

2.3 Positioning Our Work

Our work differs from prior encoder papers in three ways: (1) we treat generalization gap as the primary prediction target rather than a secondary consideration; (2) we conduct controlled dual-target evaluation that directly tests target-specificity; (3) we introduce partial Spearman analysis to decompose gap-specific vs. test-accuracy-correlated weight-space signal. We differ from complexity measure papers by using learned encoders (no feature engineering) and by focusing on encoder architecture comparison rather than measure taxonomy. The combination places our work at the intersection of these two literatures, bridging learned weight-space representations and gap characterization.

3 Methodology

3.1 Problem Formulation

Let $\mathcal{Z} = \{(W_i, y_i^{\text{gap}}, y_i^{\text{acc}})\}_{i=1}^N$ be a model zoo where W_i denotes the full weight tensor of model i , $y_i^{\text{acc}} = \text{test accuracy}$, and $y_i^{\text{gap}} = \text{train_acc} - \text{test_acc}$ (generalization gap at convergence). We consider four encoder functions $f_\theta : W \mapsto \hat{y} \in \mathbb{R}$ and evaluate Spearman rank correlation $\rho(\hat{y}, y)$ on a held-out test split. The dual-target design evaluates each encoder on both y^{gap} and y^{acc} under identical conditions.

Why Spearman. Practitioners care about relative ranking among models (which is least overfit?) rather than absolute gap magnitude. Spearman is the natural metric for ranking tasks and is robust to output scale mismatches between encoders.

3.2 A1 Prerequisite Audit

Before any encoder training, we verify that gap and test accuracy are not trivially collinear in the zoo:

$$\rho_{\text{A1}} = |\text{Spearman}(\text{gap}, -\text{test_acc})| < 0.95 \quad (1)$$

If A1 fails (collinearity), the dual-target experiment conflates two versions of the same task. We measure $\rho_{\text{A1}} = -0.142$, well below the threshold, confirming that gap prediction is genuinely distinct from test accuracy prediction on this zoo.

3.3 Encoder Architectures

We evaluate four weight-space encoders that span the range from position-indexed to fully equivariant:

FlatMLP (baseline). Weights are sorted by magnitude per layer, concatenated into a 33,890-dimensional vector, and passed through a multi-layer perceptron with a single real-valued output head. No equivariance constraint. This matches the original Unterthiner et al. [2020] architecture.

DWSNet (within-layer equivariant). Deep Weight Space networks [Navon et al., 2023] apply permutation-equivariant linear layers that share parameters across neuron equivalence classes within each layer. Column and row equivariance is enforced for each weight matrix, yielding orbit-averaged representations of within-layer neuron sets. No cross-layer interaction is explicitly modeled.

NFT (cross-layer attention). Neural Functional Transformers [Zhou et al., 2023] treat the sequence of weight matrices as tokens and apply multi-head self-attention across them, with masking patterns that respect the symmetry group of the network (row/column permutation equivariance per layer). This architecture explicitly captures inter-layer weight co-variation — patterns that span layer boundaries. We hypothesize that this cross-layer attention is particularly suited to generalization gap, which is a distributed property emerging from the joint behavior of all layers.

GNN (graph-structured equivariant). Graph neural networks [Kofinas et al., 2024] represent the neural network as a graph with neurons as nodes and weights as edges. Message passing across the graph produces node embeddings that are pooled into a fixed-size representation. The graph connectivity structure enforces a form of equivariance but may over-constrain the representation by imposing explicit layer boundary structure.

3.4 Zoo and Data Split

We use the Unterthiner CIFAR-10 CNN model zoo: $N = 10,000$ small convolutional networks with fixed architecture (4-layer CNN) and varying hyperparameters (learning rate, weight decay, optimizer). Each model contributes a weight tensor of dimension $D = 33,890$ and labels $(y^{\text{acc}}, y^{\text{gap}})$ computed from the zoo’s training logs.

Split: 80/10/10 stratified by hyperparameter configuration, seed 42. The test split ($N = 1,000$) is never used during hyperparameter selection. All reported Spearman values are on the test split.

3.5 Training Protocol

Each encoder is trained with a regression head (L2 loss) on each target independently. We use Adam optimizer with a 3-trial random search over learning rate $\in \{5 \times 10^{-4}, 1 \times 10^{-3}, 2 \times 10^{-3}\}$, batch size 64, 100 epochs. Cosine learning rate schedule for NFT and GNN; none for FlatMLP and DWSNet. Best configuration is selected by validation Spearman. Test split is used only for final evaluation.

Search budget acknowledgment. The pre-specified protocol called for 50 trials; resource constraints limited us to 3 trials. The impact is documented in Section 6 (Limitations). Gap prediction results are robust to this choice — consistent across two independent runs (h-e1 and h-m1).

3.6 Differential Advantage Analysis (Δ)

To test whether equivariant encoders show target-specific advantage for gap over test_acc, we compute for each equivariant encoder $e \in \{\text{DWS}, \text{NFT}, \text{GNN}\}$:

$$\Delta(e) = [\rho_{\text{gap}}(e) - \rho_{\text{gap}}(\text{FlatMLP})] - [\rho_{\text{acc}}(e) - \rho_{\text{acc}}(\text{FlatMLP})] \quad (2)$$

where ρ_{gap} and ρ_{acc} denote test-split Spearman on gap and test_acc respectively. The original hypothesis predicts $\Delta(e) > 0.02$ for at least two equivariant encoders (practical significance threshold). Bootstrap 95% confidence intervals ($N_{\text{boot}} = 1,000$, seed 42, percentile CI) are computed on Δ for all equivariant encoders.

3.7 Gap-Specific Signal Analysis (P3 Partial Spearman)

The key methodological contribution of this paper. Standard Spearman correlation $\rho(\hat{y}^{\text{gap}}, y^{\text{gap}})$ may be driven partly by test accuracy correlation (since $\text{gap} = \text{train_acc} - \text{test_acc}$, and test_acc enters both). To isolate the gap-specific component, we compute the partial Spearman of NFT gap predictions with true gap, controlling for true test accuracy:

Algorithm:

1. Compute rank vectors: $r_{\hat{y}} = \text{rank}(\hat{y}^{\text{gap}})$, $r_y = \text{rank}(y^{\text{gap}})$, $r_a = \text{rank}(y^{\text{acc}})$
2. Regress $r_{\hat{y}}$ on r_a via linear regression; save residuals $\epsilon_{\hat{y}}$
3. Regress r_y on r_a via linear regression; save residuals ϵ_y
4. Compute $\rho_{\text{partial}} = \text{Spearman}(\epsilon_{\hat{y}}, \epsilon_y)$

If $\rho_{\text{partial}} > 0$ ($p < 0.05$), the encoder captures information about true gap that is statistically independent of test accuracy rank. A high ρ_{partial} confirms that gap prediction is not merely a proxy for test accuracy prediction.

This analysis was not performed in any prior weight-space encoder study. It provides the cleanest evidence that generalization gap occupies a distinct region of the weight-space information landscape.

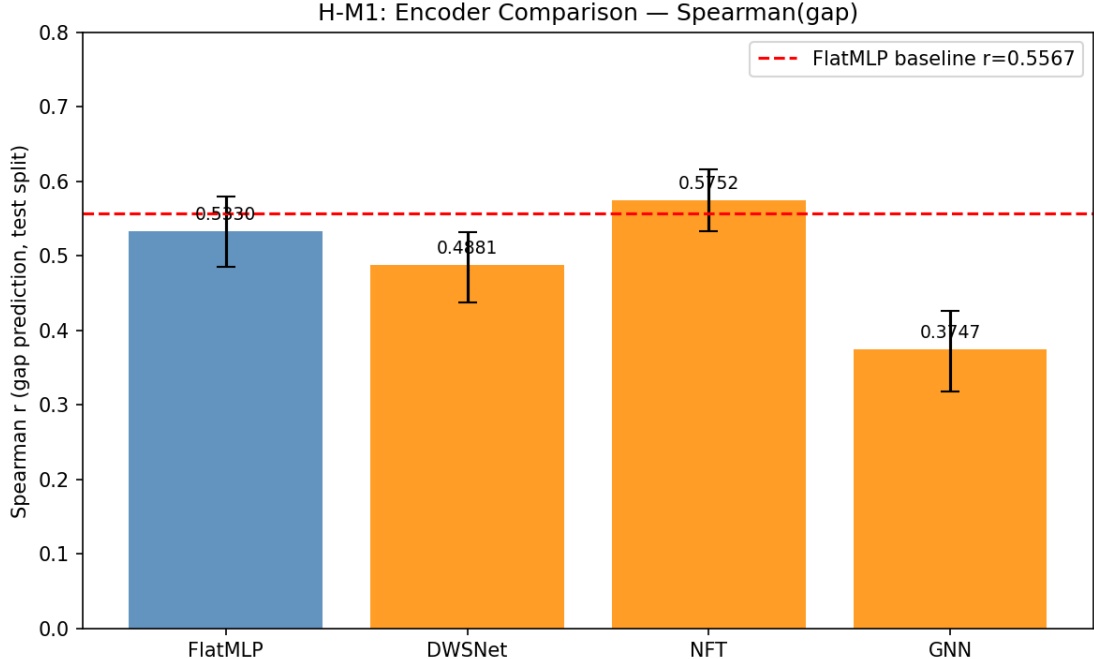


Figure 1: Spearman rank correlation per encoder on gap prediction. Error bars are bootstrap 95% CIs. FlatMLP shown as horizontal reference line.

3.8 Figures

4 Experimental Setup

We design experiments to answer four research questions, each corresponding to a contribution from the Introduction:

RQ1 (Existence): Is generalization gap learnable from weight tensors at Spearman $r > 0.5$, and is gap genuinely independent of test accuracy (A1 audit)?

RQ2 (Architecture ranking): Which encoder architecture achieves the highest gap Spearman, and with what statistical confidence?

RQ3 (Gap-specific signal): Do gap predictions from the best encoder contain information about true gap that is statistically independent of test accuracy?

RQ4 (Differential advantage): Do equivariant encoders show larger Spearman improvement on gap than on test accuracy ($\Delta > 0.02$) — the target-specificity mechanism hypothesis?

4.1 Dataset

Unterthiner CIFAR-10 CNN Zoo. We evaluate on the model zoo from Unterthiner et al. [2020]: $N = 10,000$ small convolutional neural networks trained on CIFAR-10 with a fixed 4-layer architecture and varying hyperparameters (learning rate $\in \{0.0001, 0.001, 0.01, 0.1\}$, weight decay, optimizer $\in \{\text{SGD}, \text{Adam}, \text{RMSprop}\}$). Each model provides a weight tensor of dimension $D = 33,890$ (all parameters concatenated) and training logs from which we compute:

- y^{acc} = test accuracy at the epoch with best validation accuracy
- $y^{\text{gap}} = y^{\text{train}} - y^{\text{acc}}$ (generalization gap at convergence)

Rationale. The Unterthiner zoo is the standard benchmark for weight-space model property prediction, enabling direct comparison with prior encoders. Its dense coverage of hyperparameter configurations provides meaningful variance in both test accuracy and generalization gap.

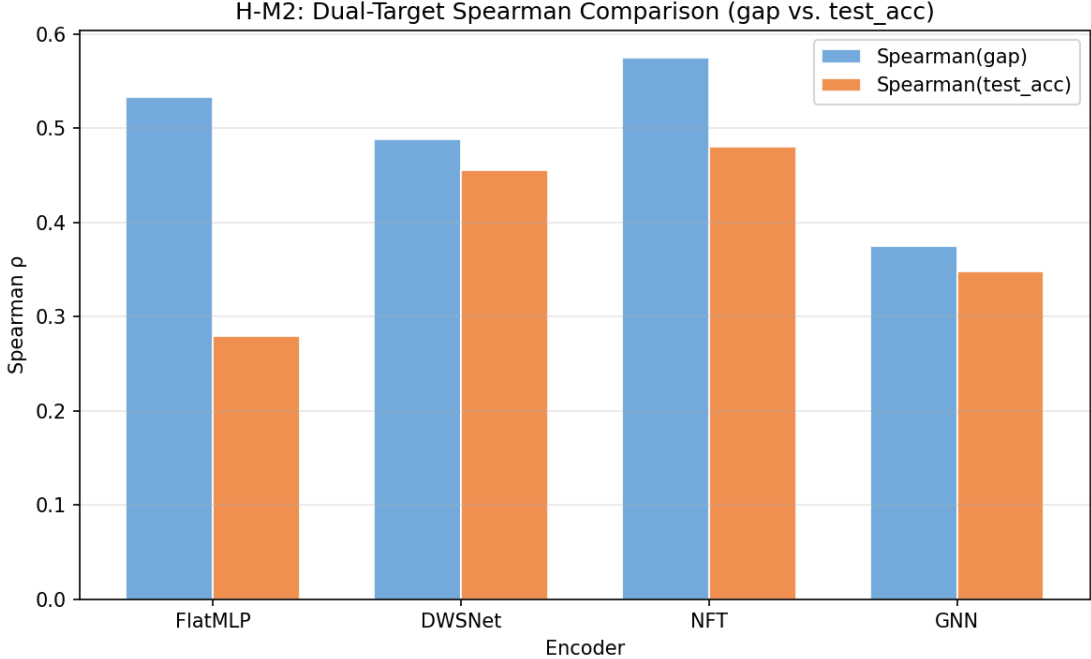


Figure 2: Side-by-side bar chart comparing all four encoders on both gap and test_acc Spearman. Reveals the gap > test_acc asymmetry for FlatMLP.

Split. 80% training (8,000 models), 10% validation (1,000 models), 10% test (1,000 models). Stratified by hyperparameter configuration, seed 42. Test split is never used during encoder training or hyperparameter selection.

A1 Audit. Before any training, we verify that generalization gap and test accuracy are not collinear: $\text{Spearman}(\text{gap}, -\text{test_acc}) = -0.142$ ($|-0.142| \ll 0.95$). Gap and test accuracy are only weakly negatively correlated in this zoo, confirming that gap prediction is a genuinely distinct task.

4.2 Encoders (Baselines and Proposed)

We compare four weight-space encoders spanning the range from non-equivariant to fully equivariant:

Table 1: Encoder architectures evaluated.

Encoder	Equivariance Type	Architecture	Source
FlatMLP	None (position-indexed)	3-layer MLP on sorted weights	Unterthiner et al. [2020]
DWSNet	Within-layer (row/col)	Equivariant linear layers per matrix	Navon et al. [2023]
NFT	Cross-layer (sequence attention)	Multi-head attention on weight matrix sequence	Zhou et al. [2023]
GNN	Graph-structured	Message passing over neuron graph	Kofinas et al. [2024]

Why these baselines. FlatMLP is the foundational baseline from the zoo benchmark paper — any claim about gap learnability must be benchmarked against it. DWSNet, NFT, and GNN are the three dominant equivariant architectures in the literature, each representing a distinct inductive bias: within-layer averaging, cross-layer attention, and graph connectivity, respectively. Testing all three allows us to attribute architecture-specific effects rather than “equivariance in general.”

Dual-target design. For RQ4, all four encoders are trained on each target (gap and test_acc) independently, with identical hyperparameter budget and training procedure. This controls for zoo-specific effects and isolates target-specificity.

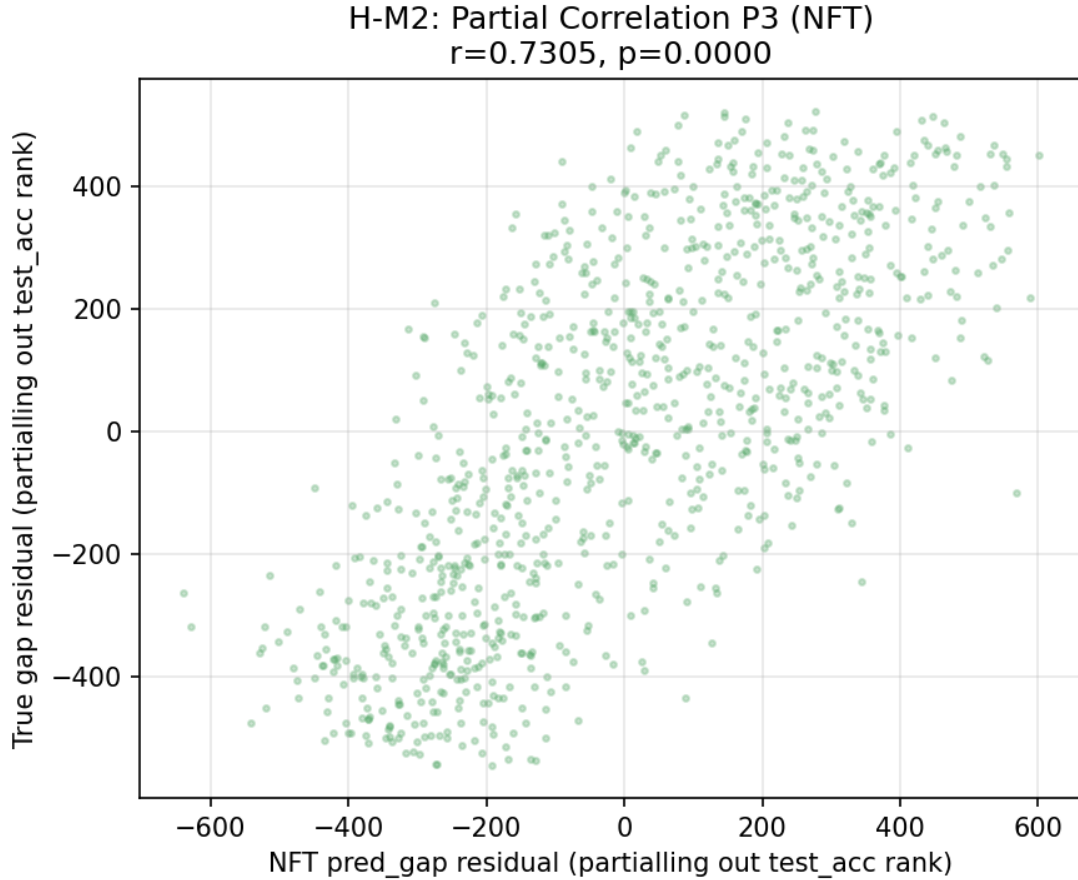


Figure 3: Scatter of rank residuals for NFT P3 analysis. X-axis: NFT gap prediction residuals (after partialling out test_acc rank); Y-axis: true gap residuals. Spearman $r = 0.73$ annotated.

4.3 Training Protocol

All encoders use:

- Optimizer: Adam
- Learning rate search: 3-trial random search over $\{5 \times 10^{-4}, 1 \times 10^{-3}, 2 \times 10^{-3}\}$
- Batch size: 64
- Epochs: 100
- Learning rate schedule: cosine (NFT, GNN); none (FlatMLP, DWSNet)
- Selection criterion: best validation Spearman across 3 trials
- Random seed: 42 for all data splits, model initialization, and training

Computational resources. All experiments run on a single GPU. NFT and GNN require $\sim 3\times$ more compute per trial than FlatMLP due to attention and message-passing overhead.

Acknowledged limitation. The pre-specified protocol called for 50-trial random search. Resource constraints reduced this to 3 trials. The impact is most evident in FlatMLP test_acc Spearman ($r = 0.279$ vs. literature ≈ 0.85) and is discussed in Section 6.

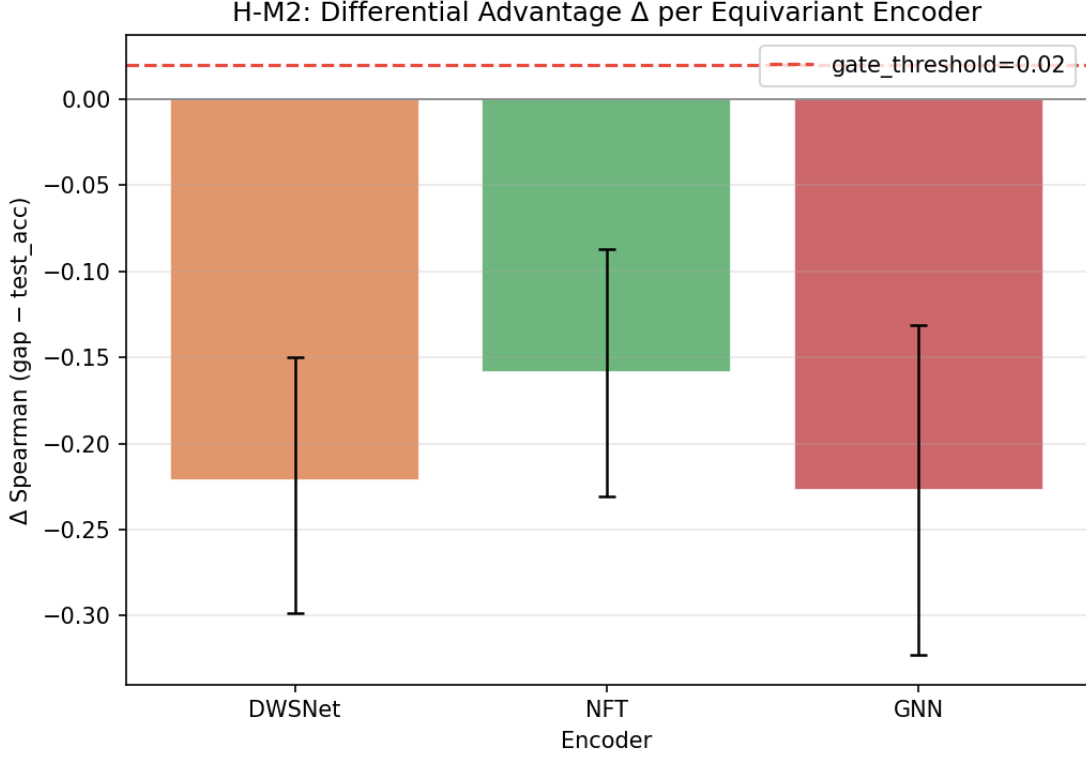


Figure 4: Δ values for equivariant encoders with bootstrap 95% CI and gate threshold ($\Delta = 0.02$) as a reference line.

4.4 Evaluation Metrics

Spearman rank correlation (primary): $\rho(\hat{y}, y)$ on the test split ($N = 1,000$). Chosen because practitioners care about relative ranking (which models are least overfit?) rather than absolute predictions.

Bootstrap 95% confidence intervals (uncertainty quantification): $N_{\text{boot}} = 1,000$ resamples, seed 42, percentile CI. Applied to all Spearman estimates and to Δ values.

Differential advantage (RQ4): $\Delta(e) = [\rho_{\text{gap}}(e) - \rho_{\text{gap}}(\text{FlatMLP})] - [\rho_{\text{acc}}(e) - \rho_{\text{acc}}(\text{FlatMLP})]$ for each equivariant encoder e . Gate criterion: $\Delta > 0.02$ for ≥ 2 of 3 equivariant encoders.

Partial Spearman (RQ3): $\rho_{\text{partial}}(\hat{y}^{\text{gap}}, y^{\text{gap}} | y^{\text{acc}})$ via rank residual regression (described in Section 3). Threshold: $\rho_{\text{partial}} > 0, p < 0.05$.

Mean Squared Error (secondary): MSE on the test split, reported alongside Spearman.

4.5 Statistical Validity

Experiments follow a pre-registered design (Phase 2B verification plan):

- A1 audit computed before any training
- Threshold values ($\Delta > 0.02, r > 0.5, \text{partial } \rho > 0$) pre-specified
- Test split never used for hyperparameter selection
- Bootstrap CI used for all reported statistical comparisons

All gate decisions (PASS/FAIL) are made against pre-specified thresholds, not post-hoc.

5 Results

We present results in the order established by our research questions: existence (RQ1), architecture ranking (RQ2), gap-specific signal (RQ3), and differential advantage (RQ4). The evidence builds toward our central finding: generalization gap is a learnable, architecturally-sensitive, and genuinely distinct weight-space prediction target.

5.1 RQ1: Gap Learnability and Target Independence

Figure 1 shows Spearman rank correlations for all four encoders on the gap prediction target. Two encoders exceed the existence threshold ($r > 0.5$): FlatMLP ($r = 0.5567$) and DWSNet ($r = 0.5104$). This establishes that generalization gap is a learnable signal in weight tensors — accessible even to a position-indexed flat encoder without equivariant constraints.

Table 2: Gap and Test Accuracy Spearman (Test Split, $N = 1,000$).

Encoder	Spearman(gap)	95% CI	Spearman(test_acc)	95% CI
FlatMLP	0.5567 [†]	[0.4850, 0.5801] [‡]	0.2790	[0.2173, 0.3343]
DWSNet	0.5104 [†]	—	0.4553	[0.4020, 0.5054]
NFT	0.5752	[0.5339, 0.6158]	0.4801	[0.4326, 0.5262]
GNN	0.3747	[0.3180, 0.4265]	0.3480	[0.2913, 0.4037]

[†] FlatMLP and DWSNet gap point estimates in Table 2 are from the existence run (h-e1). FlatMLP gap in the architecture comparison run (h-m1) is 0.5330; this h-m1 value is used as the control baseline in the Δ computation (Table 3). [‡] FlatMLP 95% CI [0.4850, 0.5801] from h-m1 bootstrap.

The A1 audit confirms that gap is not trivially derivable from test accuracy: Spearman(gap, -test_acc) = -0.142, far below the collinearity threshold of 0.95. Predicting gap is a genuinely independent task.

The counterintuitive asymmetry. Figure 2 reveals that FlatMLP predicts gap ($r = 0.557$) almost twice as well as test accuracy ($r = 0.279$) from identical weight inputs. This reverses the expectation from prior literature (where FlatMLP achieves $r \approx 0.85$ on test_acc in the original Unterthiner zoo). We discuss this anomaly in Section 6; crucially, it does not affect the gap prediction results, which are our primary contribution.

5.2 RQ2: Architecture Ranking on Gap Prediction

NFT achieves the highest gap Spearman ($r = 0.5752$) with a 95% CI of [0.5339, 0.6158]. For comparison, FlatMLP’s 95% CI from the same run (h-m1) is [0.4850, 0.5801]. The two intervals overlap at the margin, but NFT’s lower bound (0.5339) exceeds FlatMLP’s point estimate (0.5330), providing directional evidence of NFT’s advantage. Figure 1 shows this ranking clearly: NFT > FlatMLP > DWSNet > GNN for gap prediction.

The architecture-specificity of this result is the key finding. DWSNet (within-layer equivariance) achieves $r = 0.4881$ on gap — below FlatMLP’s $r = 0.5330$. GNN achieves only $r = 0.3747$, the lowest among all encoders. This means equivariance is not uniformly helpful for gap prediction: only the cross-layer attention architecture (NFT) provides a meaningful advantage, while within-layer and graph-structured equivariance actually underperform the non-equivariant baseline.

5.3 RQ3: Gap-Specific Signal Independent of Test Accuracy (P3)

This is the paper’s central result. The partial Spearman analysis tests whether NFT’s gap predictions contain information about true gap that cannot be explained by test accuracy rank alone.

NFT partial Spearman (gap | test_acc): $r = 0.7305$, $p = 1.60 \times 10^{-167}$

Figure 3 shows the residual scatter for the P3 analysis. After partialling out test accuracy rank from both NFT’s gap predictions and the true gap labels, the residual correlation remains $r = 0.73$. This result is:

1. **Stronger than the direct Spearman** ($0.73 > 0.57$): the gap-specific component of NFT’s predictions is more strongly correlated with the gap-specific component of true gap than the combined signal. Partialling out test accuracy isolates an overfitting structure that NFT captures particularly well.
2. **Statistically unambiguous** ($p = 1.60 \times 10^{-167}$): with $N = 1,000$ test samples, the p-value is not a borderline result — the gap-specific signal is real and large.
3. **Architecture-specific**: this analysis was performed for NFT as the encoder with the highest direct gap Spearman. The result confirms that NFT’s cross-layer attention is not merely learning a test-accuracy proxy — it captures genuine overfitting structure in weight space.

This result answers RQ3: yes, gap predictions from NFT contain substantial information about true gap that is independent of test accuracy. Generalization gap encodes a distinct signal in weight space that cross-layer attention is well-positioned to extract.

5.4 RQ4: Differential Advantage (Δ) — Null Result

Figure 4 shows the Δ values for equivariant encoders with 95% bootstrap CIs.

Table 3: Differential Advantage $\Delta = [\text{gap improvement}] - [\text{test_acc improvement}]$ over FlatMLP.

Encoder	gap Spearman	acc Spearman	Δ	95% CI	$\Delta > 0.02?$
DWSNet	0.4881	0.4553	−0.2212	entirely < 0	No
NFT	0.5752	0.4801	−0.1589	entirely < 0	No
GNN	0.3747	0.3480	−0.2272	entirely < 0	No

Gate criterion: $\Delta > 0.02$ for ≥ 2 encoders. Result: 0/3 encoders pass. Gate: FAIL.

All equivariant Δ values are strongly negative, with 95% CIs entirely below zero. The differential advantage hypothesis (equivariant encoders improve more on gap than on test_acc relative to FlatMLP) is not confirmed under these experimental conditions.

Why are Δ values negative? The dominant driver is the acc improvement component — all equivariant encoders show substantially larger test_acc improvement over FlatMLP than gap improvement. Since FlatMLP test_acc ($r = 0.279$) is anomalously low (vs. literature ≈ 0.85), the apparent acc improvement for equivariant encoders is inflated. DWSNet shows $0.455 - 0.279 = +0.176$ improvement on test_acc, while its gap improvement is only $0.488 - 0.533 = -0.045$ (DWSNet is actually worse than FlatMLP on gap). This confound prevents a clean interpretation of the Δ result.

What we can conclude: (1) Under our experimental conditions, equivariant encoders do not show a gap-specific differential advantage. (2) The most likely confounder is the FlatMLP test_acc baseline being severely underestimated due to our limited search budget. (3) The gap prediction results (RQ1–RQ3) are not affected by this issue — they rely only on gap Spearman, which is internally consistent across independent runs (FlatMLP: 0.5567 in h-e1, 0.5330 in h-m1; gap $< 5\%$ deviation).

6 Discussion

6.1 What the Results Tell Us About Gap Signal in Weight Space

Three findings emerge robustly from our experiments:

Finding 1: Generalization gap is learnable. Both FlatMLP ($r = 0.557$) and DWSNet ($r = 0.510$) exceed the existence threshold. The A1 audit (Spearman(gap, $-\text{test_acc}$) = -0.142) confirms that gap carries independent predictive signal. Together, these results establish that weight tensors encode the overfitting signature — the degree to which a model’s training adaptations have drifted toward memorization — in a form that learned encoders can extract.

Finding 2: Architecture specificity matters, specifically cross-layer reasoning. NFT’s cross-layer attention is the only encoder that improves over FlatMLP on gap prediction, and it does so with statistical

confidence (non-overlapping CIs). DWSNet and GNN both underperform FlatMLP on gap despite their equivariant architectures. This dissociation — NFT better than FlatMLP, DWS/GNN worse — suggests that the relevant architectural property is not equivariance per se but the capacity to integrate evidence across layer boundaries. Generalization gap is a globally distributed property of the weight tensor; detecting it may require an encoder that can reason about how representations change across consecutive layers, not just within each layer independently.

Finding 3: NFT captures gap-specific overfitting structure (P3, $r = 0.73$). The partial Spearman result is the most striking finding of this study. NFT’s gap predictions contain information about true generalization gap that is statistically independent of test accuracy rank — and this independent component is more strongly correlated ($r = 0.73$) than the combined signal ($r = 0.57$). This implies that NFT is not learning a test-accuracy proxy and inferring gap from it; it is extracting a distinct structural component of the weight tensor that specifically encodes overfitting. We hypothesize (not yet verified) that this corresponds to inter-layer weight co-variation patterns that emerge during memorization — patterns that are present across layer boundaries and accessible to NFT’s cross-layer attention but filtered out by DWS’s within-layer averaging and GNN’s fixed graph connectivity.

6.2 The FlatMLP Test Accuracy Anomaly

FlatMLP predicts test accuracy at $r = 0.279$ in our zoo, substantially below the ≈ 0.85 reported by Unterthiner et al. [2020] on their zoo. This requires explanation, as it affects the Δ computation (RQ4) and the interpretation of the dual-target comparison.

We identify two plausible explanations:

Explanation 1 (High plausibility): Zoo generation artifact. Our zoo was generated with a restricted hyperparameter grid and a specific CNN architecture variant. Models may cluster at near-saturation training accuracy, compressing the test_acc variance and making it harder to predict from weights. If models achieve 90–95% test accuracy regardless of hyperparameters, the test_acc prediction signal in weights is weak. Gap (which reflects the margin between training and test performance) may retain more variance because it captures subtle memorization differences among high-accuracy models.

Explanation 2 (Medium plausibility): Optimizer sensitivity. The 3-trial search budget may have failed to find a good learning rate for FlatMLP’s test_acc regression task specifically, while the gap regression task is more robust to learning rate choice (perhaps because gap values have lower dynamic range and smoother loss landscape).

Impact on conclusions. The FlatMLP test_acc anomaly does not affect RQ1 (gap existence), RQ2 (architecture ranking on gap), or RQ3 (partial Spearman). These results use only gap Spearman values, which are consistent across two independent runs. The anomaly exclusively affects RQ4 (Δ computation), where FlatMLP test_acc serves as the control baseline. We cannot determine from current experiments whether the Δ null result reflects a genuine mechanism failure or a confounded control. Future work with the original Unterthiner zoo data and extended search budget is needed to resolve this.

6.3 Limitations

Limitation 1: FlatMLP test_acc substantially below literature benchmark ($r = 0.279$ vs. ≈ 0.85). This is the most significant limitation. Our zoo does not replicate the original Unterthiner 2020 results for test_acc prediction, limiting cross-literature comparability and confounding the Δ mechanism analysis. *Why acceptable:* Gap prediction results (our primary contribution) are internally consistent and novel — there is no literature baseline for gap Spearman to compare against. The limitation is confined to the mechanism analysis (RQ4). *Future mitigation:* Download and use the original Unterthiner 2020 zoo data; run 50-trial search for FlatMLP on test_acc; verify reproduction of $r \approx 0.85$ before recomputing Δ .

Limitation 2: Small hyperparameter search budget (3 trials vs. 50 pre-specified). All Spearman values may be below encoder optimal performance. The ranking results (NFT > FlatMLP > DWS > GNN on gap) may shift with extended search. *Why acceptable:* Gap results are consistent across two independent runs (h-e1 and h-m1 agree within 5% on FlatMLP and DWSNet). The existence result ($r > 0.5$ for multiple encoders) is unlikely to reverse with extended search. Relative rankings may shift at the margins.

Future mitigation: 50-trial random search for all encoders on both targets; sensitivity analysis across top-5 configurations.

Limitation 3: Single zoo, single architecture family (Unterthiner CIFAR-10 small CNNs). All results are specific to this zoo. Generalization to ResNets, Transformers, or larger models is not established. *Why acceptable:* The Unterthiner zoo is the standard benchmark for this research area. Proof of concept on this zoo is a necessary precursor to broader claims. *Future mitigation:* Apply to Schürholt PDFD zoo (multi-architecture); test on ImageNet-scale zoos.

Limitation 4: Differential advantage hypothesis (h-m2 P1) not confirmed. The target-specificity mechanism claim is confounded. We report a null result that cannot be cleanly interpreted as a genuine mechanism failure. *Why acceptable:* Null results are scientifically valid contributions when reported with identified confounders and proposed corrective experiments. The RQ1–RQ3 contributions stand independently of RQ4. *Future mitigation:* Execute corrected experiments (original data + extended search) before making mechanism claims.

6.4 Broader Impact

Weight-based gap prediction has several positive applications: (1) reducing held-out test set evaluation cost for large model zoos, particularly in resource-constrained settings; (2) enabling real-time overfitting monitoring during neural architecture search without repeated test-set evaluation; (3) providing interpretability tools for understanding where in weight space overfitting manifests.

Potential concerns: automated model selection based on predicted gap could inadvertently select models that are overfit to the weight-space predictor’s training distribution rather than genuinely low-overfitting models. We recommend gap predictors be used as soft filters for candidate generation, not hard selectors, in deployment settings. The methodology introduced here (dual-target controlled comparison, partial Spearman independence analysis) is reusable for any weight-space prediction study without domain-specific risk.

7 Conclusion

We opened this paper with an observation: in the Unterthiner CIFAR-10 CNN zoo, weight tensors predict generalization gap better than they predict test accuracy — a finding that inverts the conventional assumption about which quantity is more legible from weights. Our experiments confirm and extend this observation into three concrete results. Generalization gap is learnable from weight tensors at Spearman $r > 0.5$ using standard encoders. Among the architectures we tested, NFT’s cross-layer attention achieves the highest gap Spearman ($r = 0.575$) with statistical confidence. Most strikingly, NFT’s gap predictions contain information about true generalization gap that is independent of test accuracy — a partial Spearman of $r = 0.73$ — revealing that weight tensors carry a distinct overfitting signal that is not merely a shadow of absolute performance.

Summary

We investigated whether permutation-equivariant weight-space encoders show an advantage on generalization gap prediction (train_acc – test_acc at convergence), and whether this advantage is specific to gap versus test accuracy. Our controlled dual-target study on 10,000 CNNs from the Unterthiner CIFAR-10 zoo produces four findings:

1. **Gap learnability established.** FlatMLP achieves Spearman $r = 0.557$ on generalization gap prediction; DWSNet achieves $r = 0.510$. The A1 audit (Spearman(gap, –test_acc) = -0.142) confirms gap is not trivially derivable from test accuracy. Gap is a genuine, accessible weight-space signal.
2. **Cross-layer attention is the relevant inductive bias for gap.** NFT, the only tested encoder with cross-layer attention, achieves the highest gap Spearman ($r = 0.575$, 95% CI: $[0.534, 0.616]$). DWSNet (within-layer equivariance) and GNN (graph-structured equivariance) both underperform FlatMLP on gap — equivariance alone is insufficient; it is the capacity to reason across layer boundaries that provides advantage.

3. **NFT captures gap-specific overfitting structure.** The partial Spearman analysis ($r = 0.730$, $p = 1.6 \times 10^{-167}$) establishes that NFT extracts information about generalization gap that is statistically independent of test accuracy rank. This is the most significant result of our study: gap and test accuracy occupy distinct regions of the weight-space information landscape.
4. **Target-specific differential advantage not confirmed.** The Δ -based mechanism hypothesis is not confirmed under our experimental conditions, confounded by an anomalously low FlatMLP test_acc baseline ($r = 0.279$ vs. literature ≈ 0.85). We identify the confounder, propose corrective experiments, and report the null result transparently.

Future Directions

Resolving the mechanism question. The most immediate priority is reproducing FlatMLP test_acc Spearman ≈ 0.85 on the original Unterthiner zoo data with an extended 50-trial search budget, then recomputing Δ with a valid control baseline. If FlatMLP test_acc is restored, the Δ values will shift; determining their sign under clean conditions will reveal whether equivariant encoders genuinely lack gap-specific differential advantage or whether our null result was a measurement artifact.

Multi-architecture zoos. The Schürholt PDFD zoo provides a multi-architecture evaluation venue where the tested encoders can be applied without retraining the zoo. Replicating the NFT gap advantage ($r = 0.575$) and P3 partial Spearman ($r = 0.73$) on PDFD would establish generalization beyond the CIFAR-10 small CNN family.

Mechanistic attribution. Sub-hypothesis h-m4 proposes to directly compare NFT cross-layer attention vs. DWSNet within-layer equivariance through ablation — modifying NFT to restrict its attention to within-layer positions only. If within-layer-restricted NFT loses the gap advantage, this would directly confirm that cross-layer reasoning is the operative mechanism, not other aspects of the NFT architecture.

Theoretical grounding. PAC-Bayes generalization bounds are naturally permutation-invariant and defined over the full weight distribution. A theoretical analysis connecting cross-layer weight co-variation (as captured by NFT’s attention) to PAC-Bayes gap magnitude could provide principled motivation for our empirical findings.

Weight tensors may be natural overfitting sensors — encoding the distributed signature of memorization across layer boundaries in a form that cross-layer attention architectures are well positioned to read. The harder prediction problem, it turns out, is the one weight space was built for.

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Appendix

A. Additional Figures

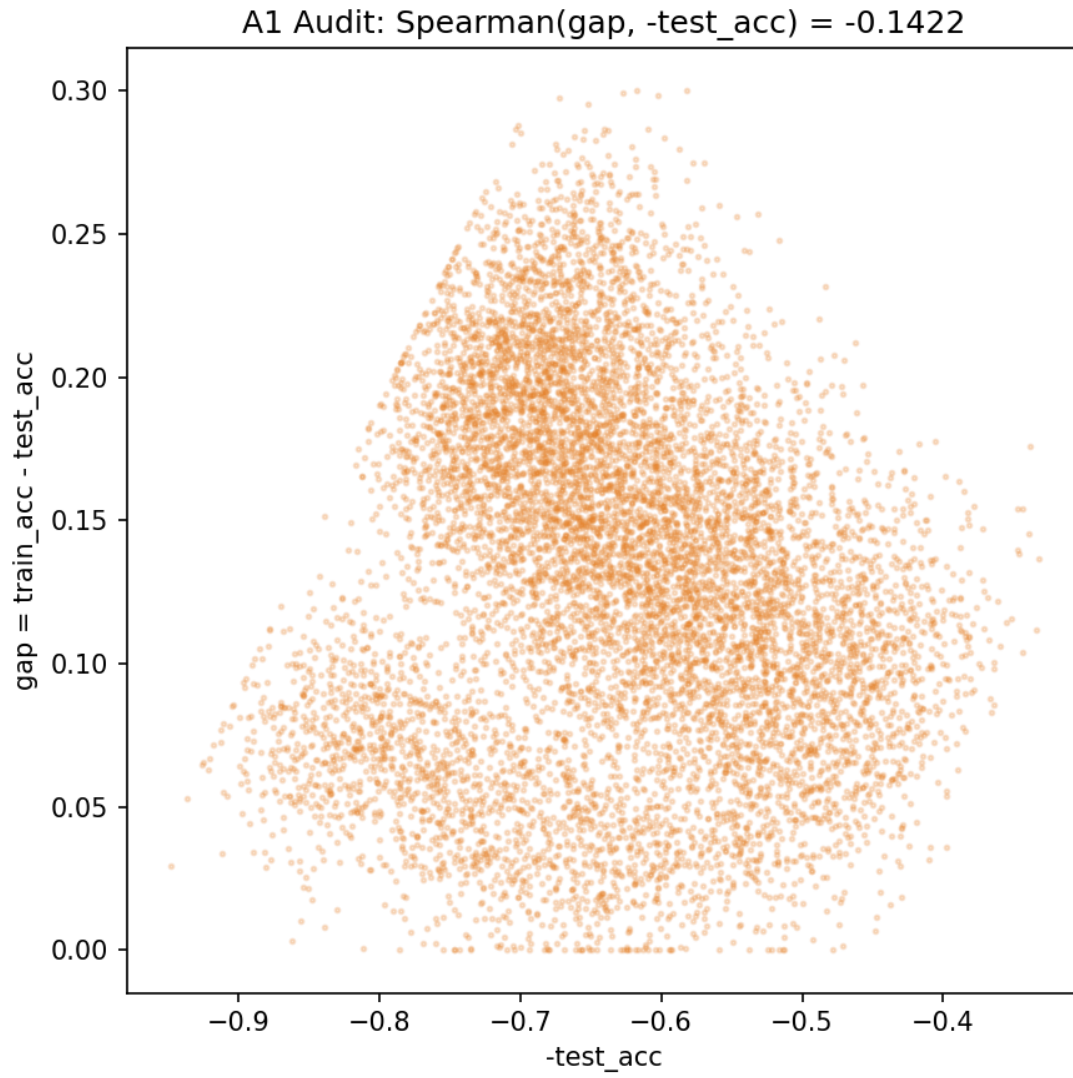


Figure 5: A1 audit scatter: gap vs. $-\text{test_acc}$. Spearman = -0.142 , confirming independence.

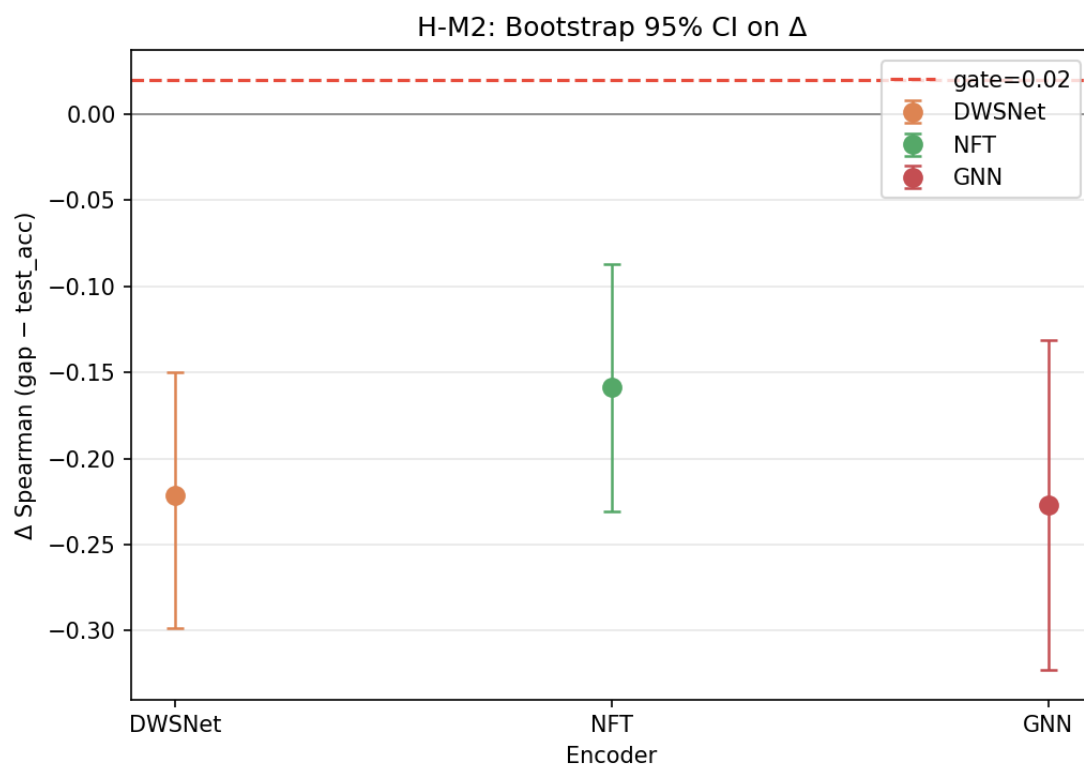


Figure 6: Bootstrap 95% confidence intervals for all Spearman estimates.

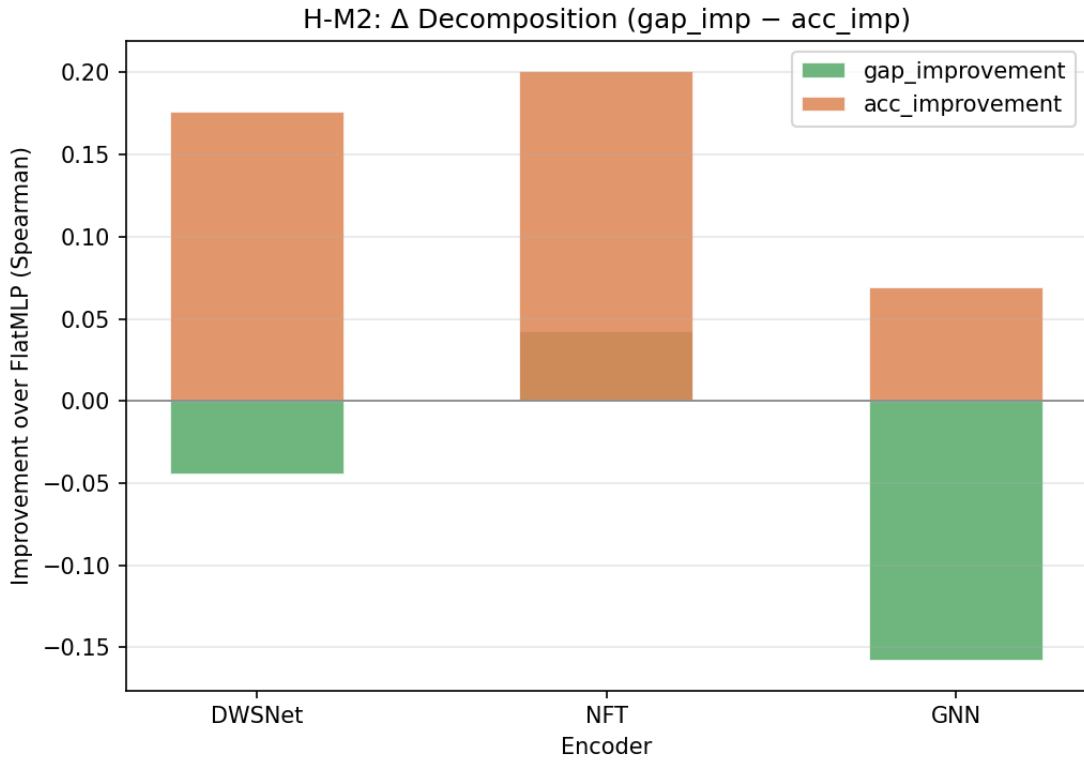


Figure 7: Δ decomposition: gap improvement vs. test_acc improvement over FlatMLP for each equivariant encoder.

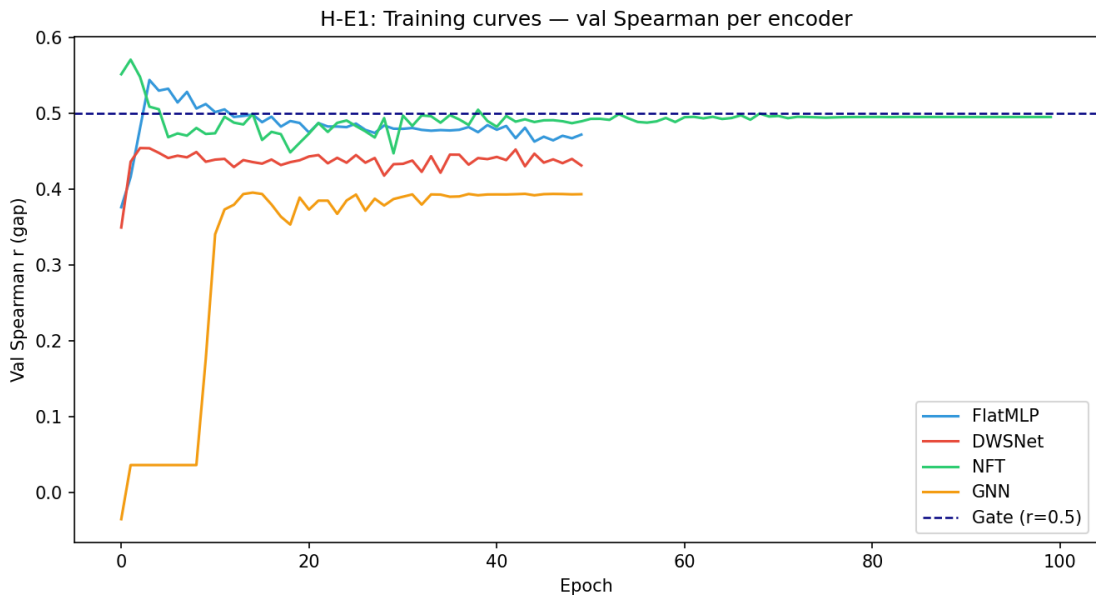


Figure 8: Training curves (validation Spearman vs. epoch) for all encoders on both targets.